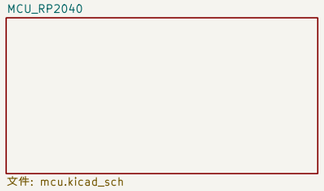
Power



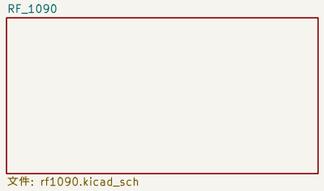
文件: power.kicad_sch

MCU_RP2040



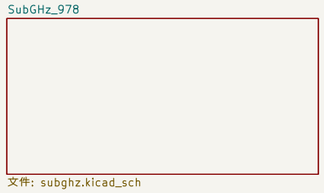
文件: mcu.kicad_sch

RF_1090



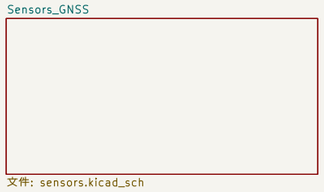
文件: rf1090.kicad_sch

SubGHz_978



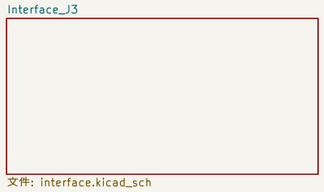
文件: subghz.kicad_sch

Sensors_GNSS



文件: sensors.kicad_sch

Interface_J3



文件: interface.kicad_sch

RF_VHF



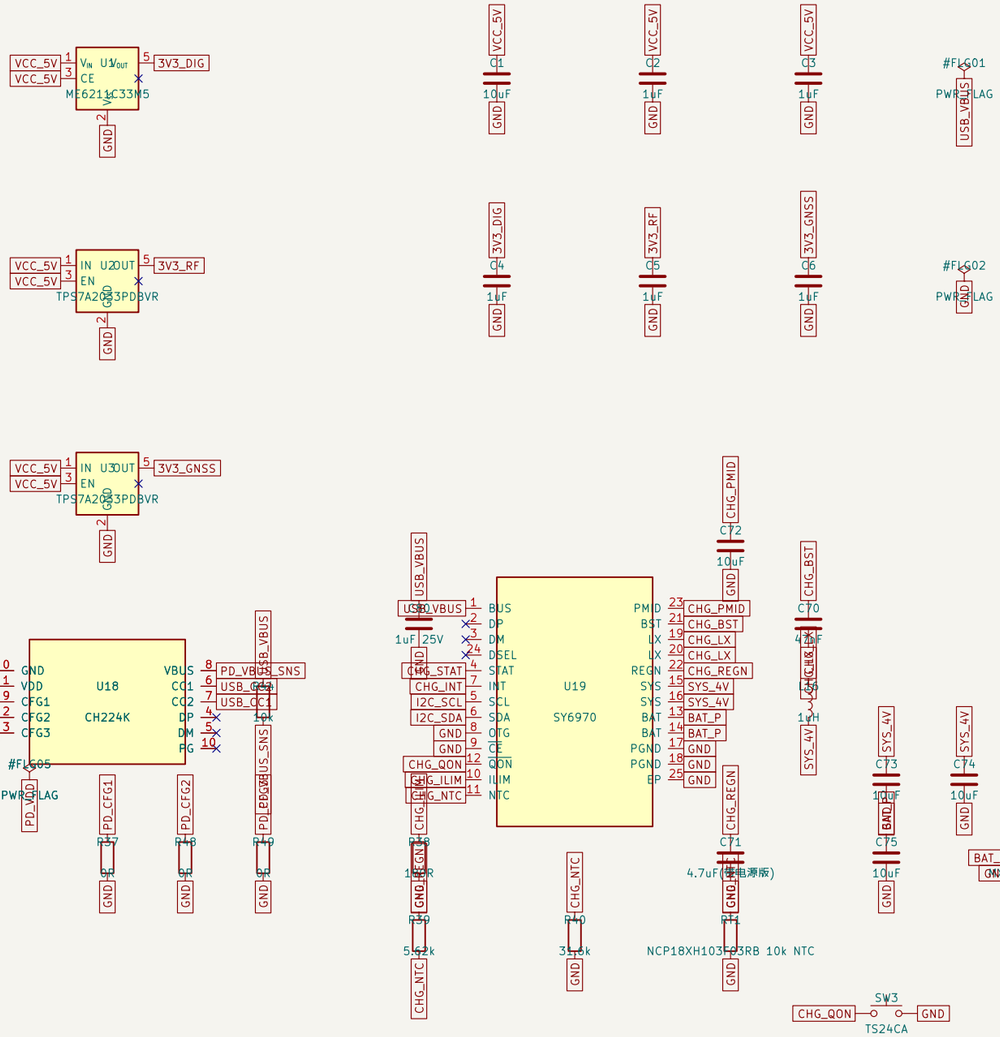
文件: rfvhf.kicad_sch

Topology: independently transcribed, no third-party design files reused
Dual-band ADS-B (1090ES/978UAT) + IMU + BARO + GNSS
Pilot Kit

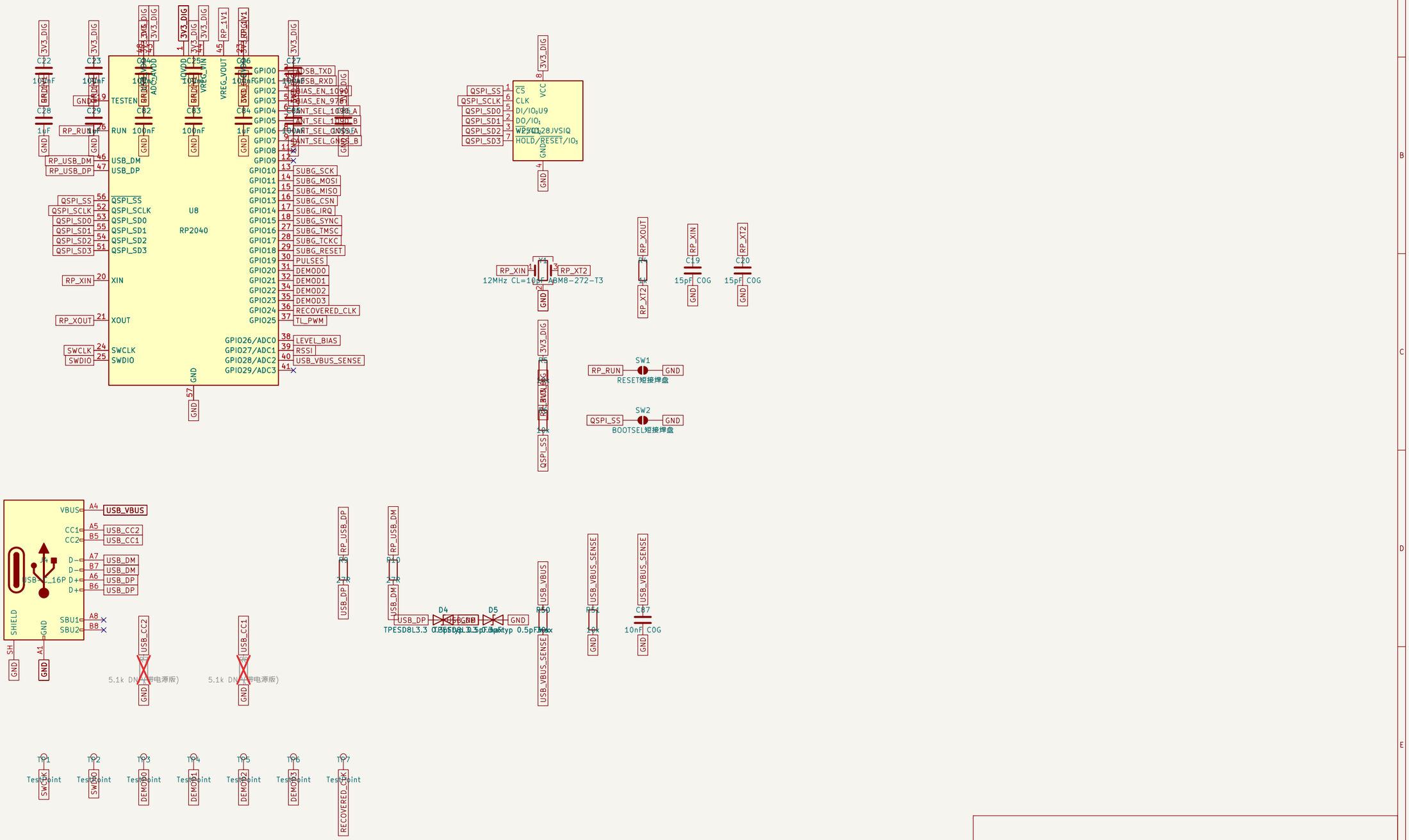
Sheet: /
File: expansion-board-v4.kicad_sch

Title: Pilot Kit Expansion Board V4.7 – Root

Size: A3	Date: 2026-08-01	**Rev: V4.7**
KiCad E.D.A. 10.0.4		Id: 1/8



Pilot Kit		
Sheet: /Power/		
File: power.kicad_sch		
Title: Expansion Board V4.7 – Power: 5V in, 3x LDO		
Size: A3	Date: 2026-08-01	**Rev: V4.7**
KiCad E.D.A. 10.0.4		Id: 2/8



Pilot Kit

Sheet: /MCU_RP2040/
File: mcu.kicad_sch

Title: Expansion Board V4.7 — MCU RP2040 + Flash + USB

Size: A3	Date: 2026-08-01	**Rev: V4.7**
KiCad E.D.A. 10.0.4		Id: 3/8

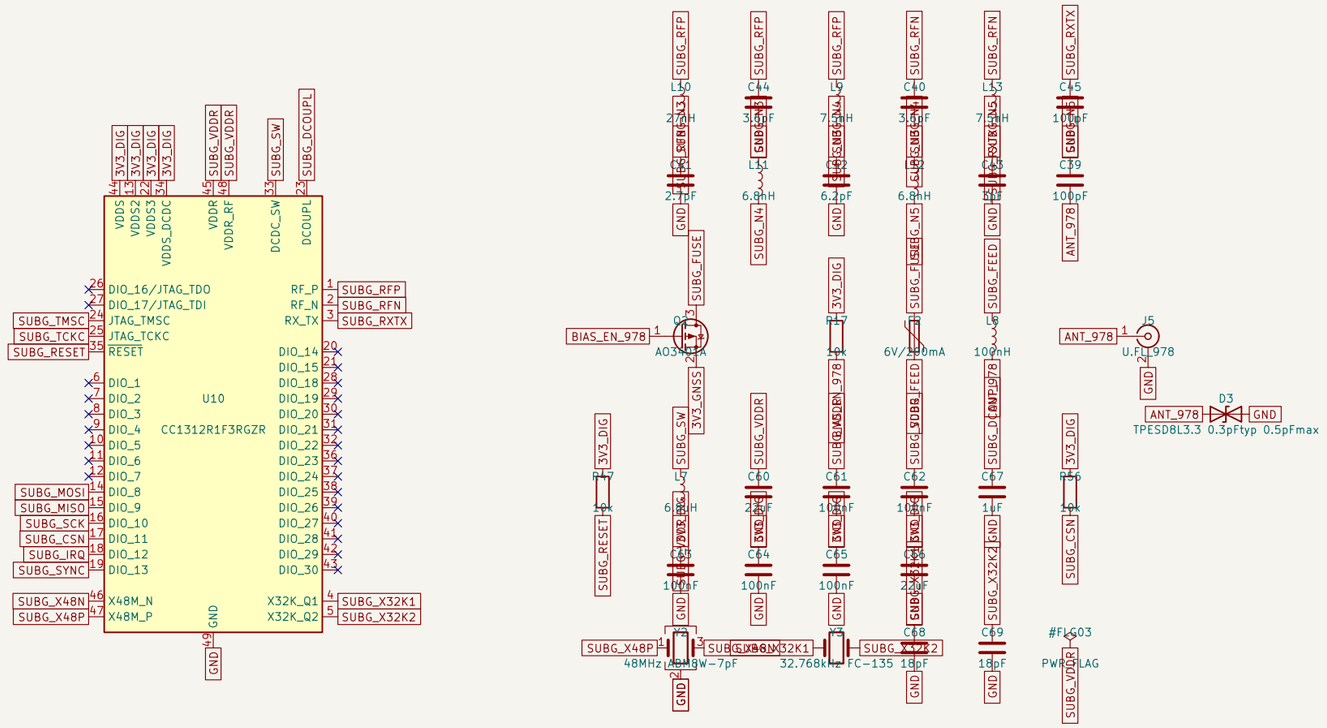
ANT1090_EXT
J6
U.FL_1090_EXT(经座接收端)
ANT1090_EXT
D2
0.3pf typ 0.5pf max
C30
100pF
ANT1090_EXT
GND
ANT1
ANT1090_IFA
GND
IFA_1090
Z?1
DNP 并上一级侧
GND
Z?1
ANT1090_EXT
GND
Z?1
DNP 并上一台侧
GND
ANT1090_EXT
RF1090_FUSE
3V3_GNSS
BIAS_EN_1090
ANT_SEL_1090_V1
ANT_SEL_1090_V2
IFA_MATCH
100pF
GND
IFA_MATCH
J1
U.FL_IFA_TEST(π后调试口)
3V3_DIG
R18
10k
RF1090_FUSE
6V/120mA
L13
RF1090_FEED
100nH
L14
RF1090_FEED
ANT1090_EXT
ANT1090_LNA
ANT1090_LNA
ANT1090_EXT
ANT1090_SW
ANT1090_SW
SW1_J1
SW1_J2
SW1_J3
J3
J2
U16
ANT_SEL_1090_V1
ANT_SEL_1090_V2
GND
R22
0R
R23
0R
C54
100pF
LNA1_IN
ANT_SEL_1090_V1
C55
100pF
GND
ANT_SEL_1090_V2
C56
100pF COG
ANT_SEL_1090_V2
SW1_OUT
SW1_J1
SW1_J2
SW1_J3
OR DNP(路外接)DNP(上路板载)
LNA1_OUT
C31
100pF COG
LNA1_IN
SAW1_IN
3V3_RF
GND
ANT_SEL_1090_V1
ANT_SEL_1090_V2
GND
ANT_FL1
A0970A
IN
OUT
GND
GND
SAW1_OUT
SAW1_IN
LNA2_IN
LNA2_OUT
C32
100pF COG
LNA2_IN
3V3_RF
GND
U12
B6A2817
Vcc
LNA2_IN
LNA2_OUT
C61
470pF COG
GND
LNA2_OUT
C33
100pF COG
SAW2_IN
GND
GND
FL2
TA0970A
IN
GND
GND
OUT
GND
GND
SAW2_OUT
C34
100pF COG
SAW2_OUT
DET_IN
R19
51.1R
DET_INLO
DET_INLO
C35
100pF COG
GND
LNA1_IN
U11
QPG6N07TR7
GND
GND
LNA1_OUT
LNA1_VBIAS
18nH 0402CS-18NXGRW
LNA1_OUT
LNA1_VBIAS
R11
2k
3V3_RF
GND
#FLG04
PWR_FLAG
GND
C36
100pF
GND
C48
1uF
GND
RF_DET_OUT
1K DNP
3V3_RF
R31
SLICER_FAST
C46
30pF
GND
R32
10k
SLICER_LEVEL
C47
200pF
GND
R33
100k
SLICER_LEVEL
R34
10k
LEVEL_BIAS
C49
100nF
GND
R35
10k
RSSI
C51
1nF
GND
RF_DET_OUT
R31
SLICER_FAST
SLICER_FAST
R32
SLICER_LEVEL
SLICER_LEVEL
R33
LEVEL_BIAS
LEVEL_BIAS
R34
TL_PWM
LEVEL_BIAS
R35
RSSI
RF_DET_OUT
C51
RSSI
AD8313_VPOS
R?4
3V3_RF
#FLG06
PWR_FLAG
C37
100nF
GND
AD8313_VPOS
C38
100pF
GND
AD8313_VPOS
R?4
AD8313_VOUT
0R
RF_DET_OUT
AD8313_VPOS
AD8313_VPOS
U14
AD8313ARMZ
VPOS
VPOS
COMM
VOUT
VSET
INHI
INLO
PWDN
DET_IN
DET_INLO
GND
GND
AD8313_VOUT
AD8313_VOUT
SLICER_FAST
SLICER_LEVEL
V+
V-
U15
LV358(TOKMAS)
SHDN
PULSES_RAW
3V3_RF
GND
GND
PULSES_RAW
R37
3.3R
PULSES
PULSES
R38
1K DNP
3V3_DIG
3V3_RF
C?2
100nF
GND

Pilot Kit

Sheet: /RF_1090/
File: rf1090.kicad_sch

Title: Expansion Board V4.7 — 1090MHz: antenna switch + RX chain

Size: A3	Date: 2026-08-01	**Rev: V4.7**
KiCad E.D.A. 10.0.4		Id: 4/8

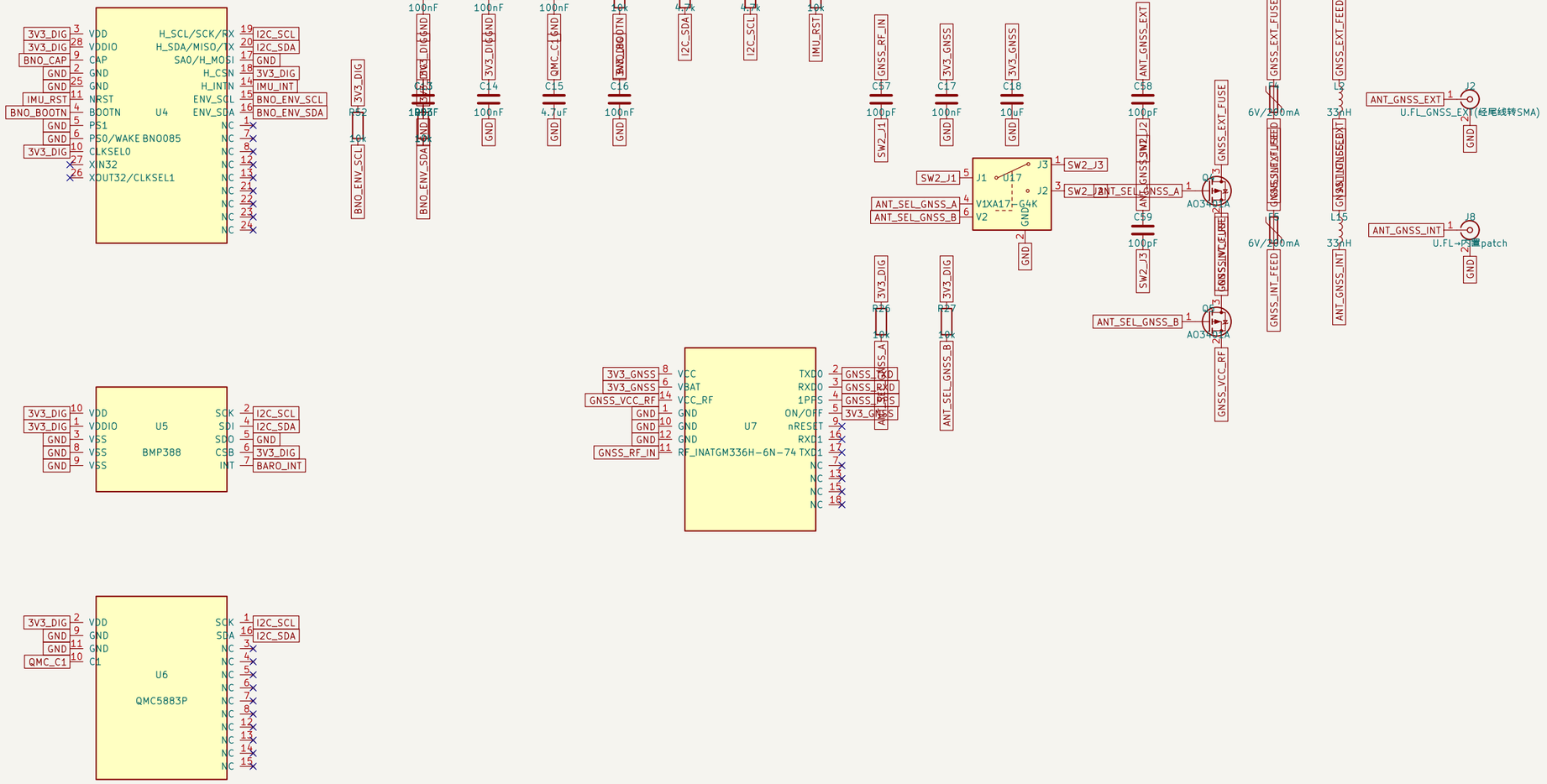


Pilot Kit

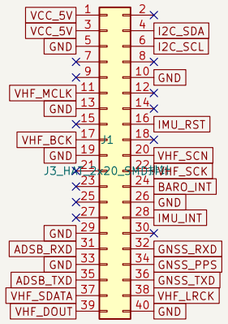
Sheet: /SubGHz_978/
File: subghz.kicad_sch

Title: Expansion Board V4.7 – 978MHz UAT: CC1312R

Size: A3	Date: 2026-08-01	**Rev: V4.7**
KiCad E.D.A. 10.0.4		Id: 5/8



Pilot Kit		
Sheet: /Sensors_GNSS/		
File: sensors.kicad_sch		
Title: Expansion Board V4.7 – Sensors + GNSS		
Size: A3	Date: 2026-08-01	**Rev: V4.7**
KiCad E.D.A. 10.0.4		Id: 6/8

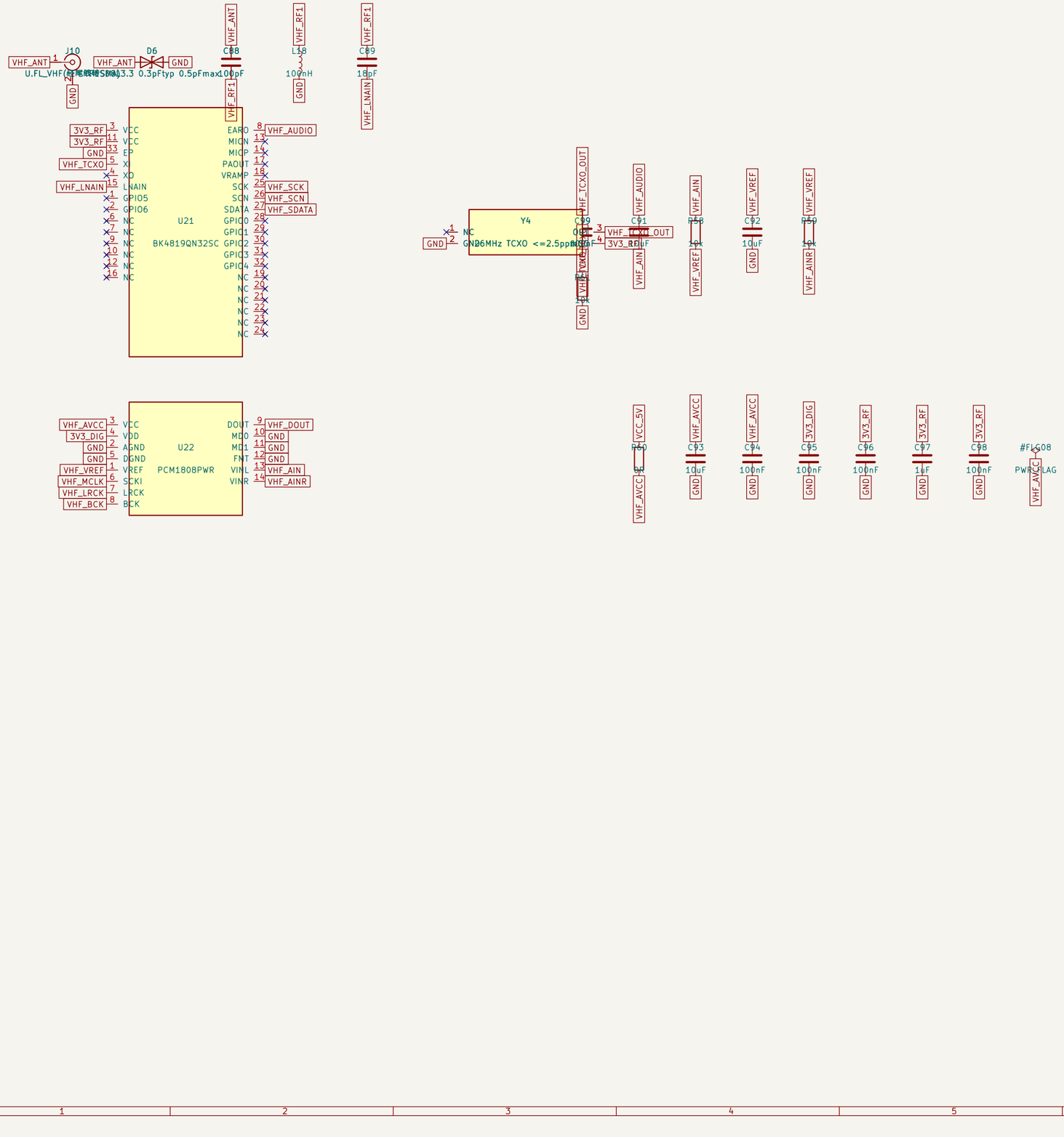


Pilot Kit

Sheet: /Interface_J3/
File: interface.kicad_sch

Title: Expansion Board V4.7 – Interface: J3 2x20 HAT socket

Size: A3	Date: 2026-08-01	**Rev: V4.7**
KiCad E.D.A. 10.0.4		Id: 7/8



Pilot Kit

Sheet: /RF_VHF/
File: rfvhf.kicad_sch

Title: Expansion Board V4.7 – VHF airband: BK4819 RX + PCM1808 capture

Size: A3	Date: 2026-08-01	**Rev: V4.7**
KiCad E.D.A. 10.0.4		Id: 8/8